

***Pandas Python library code:**

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

Load dataset

```
df = pd.read_excel("DataAnalyst_Assesment_Dataset.xlsx")
```

Drop irrelevant column

```
df.drop(columns=['Subscription Type'], inplace=True)
```

Fill missing values

```
df['Class Type'].fillna('Unknown', inplace=True)
```

```
df['Instructor'].fillna('Unknown', inplace=True)
```

```
df['Time Slot'].fillna('Not Assigned', inplace=True)
```

```
df['Duration (mins)'].fillna(df['Duration (mins)'].median(), inplace=True)
```

```
df['Facility'].fillna('Not Specified', inplace=True)
```

```
df['Theme'].fillna('Not Specified', inplace=True)
```

```
df['Customer Email'].fillna('Unknown', inplace=True)
```

```
df['Customer Phone'].fillna('Unknown', inplace=True)
```

Remove duplicates

```
df.drop_duplicates(inplace=True)
```

Convert Booking Date to datetime

```
df['Booking Date'] = pd.to_datetime(df['Booking Date'])
```

```
df['YearMonth'] = df['Booking Date'].dt.to_period('M')
```

Booking trends

```
df.groupby('YearMonth').size().plot(marker='o', linestyle='-', figsize=(12, 6), title='Monthly Booking Trends')

plt.xticks(rotation=45)

plt.show()
```

Top services

```
top_services = df.groupby('Service Name').agg(Bookings=('Booking ID', 'count'), Revenue=('Price', 'sum'))

top_services = top_services.sort_values(by='Bookings', ascending=False).head(10)
```

```
sns.barplot(x=top_services.index, y=top_services['Bookings'], palette='viridis')

plt.xticks(rotation=45)

plt.title('Top 10 Most Booked Services')

plt.show()
```

Customer Segmentation

```
customer_activity = df['Customer ID'].value_counts()

df['Customer Type'] = df['Customer ID'].map(lambda x: "Frequent" if customer_activity[x] > 5 else "Occasional")

customer_segments = df['Customer Type'].value_counts()

sns.barplot(x=customer_segments.index, y=customer_segments.values, palette='coolwarm')

plt.title('Customer Segmentation')

plt.show()
```

Revenue Trends

```
df.groupby('YearMonth')['Price'].sum().plot(marker='o', linestyle='-', figsize=(12, 6), title='Monthly Revenue Trends', color='g')

plt.xticks(rotation=45)

plt.show()
```

```
# Save cleaned dataset
```

```
df.to_csv("Cleaned_Dataset.csv", index=False)
```

***SQL query code:**

-- 1. Total bookings per month

```
SELECT DATE_FORMAT(`Booking Date`, '%Y-%m') AS YearMonth, COUNT(*) AS Total_Bookings
FROM bookings
GROUP BY YearMonth
ORDER BY YearMonth;
```

-- 2. Top 10 most booked services

```
SELECT `Service Name`, COUNT(`Booking ID`) AS Total_Bookings, SUM(Price) AS Total_Revenue
FROM bookings
GROUP BY `Service Name`
ORDER BY Total_Bookings DESC
LIMIT 10;
```

-- 3. Customer segmentation

```
SELECT `Customer ID`,
       COUNT(`Booking ID`) AS Booking_Count,
       CASE WHEN COUNT(`Booking ID`) > 5 THEN 'Frequent' ELSE 'Occasional' END AS Customer_Type
FROM bookings
GROUP BY `Customer ID`;
```

-- 4. Monthly revenue trends

```
SELECT DATE_FORMAT(`Booking Date`, '%Y-%m') AS YearMonth, SUM(Price) AS Total_Revenue
FROM bookings
GROUP BY YearMonth
ORDER BY YearMonth;
```

*Data cleaning process:

*Removed the 'Subscription Type' column (100% missing values).

*Handled missing values:

*Filled 'Class Type' & 'Instructor' with 'Unknown'.

*Filled 'Time Slot' with 'Not Assigned'.

*Replaced missing 'Duration' values with the median.

*Set missing 'Facility' & 'Theme' to 'Not Specified'.

*Replaced missing customer contact details with 'Unknown'.

*Removed duplicate rows.

*Converted 'Booking Date' to datetime format.

*EDA process:

1.Booking Trends:

Bookings fluctuate monthly, indicating seasonal demand.

Identifying peak months can optimize promotions.

2.Top Services & Revenue:

Play Area & Party Room are the most booked services.

These services also generate the highest revenue.

3.Customer Segmentation:

Most customers are Occasional (single/few bookings).

No strong customer loyalty observed.

4.Revenue Trends:

Revenue follows a similar trend to bookings.

Seasonal promotions could boost revenue in slow months.

***Business Insights & Recommendations:**

Loyalty Program: Encourage frequent bookings through discounts.

Peak Season Strategy: Increase prices or create premium packages.

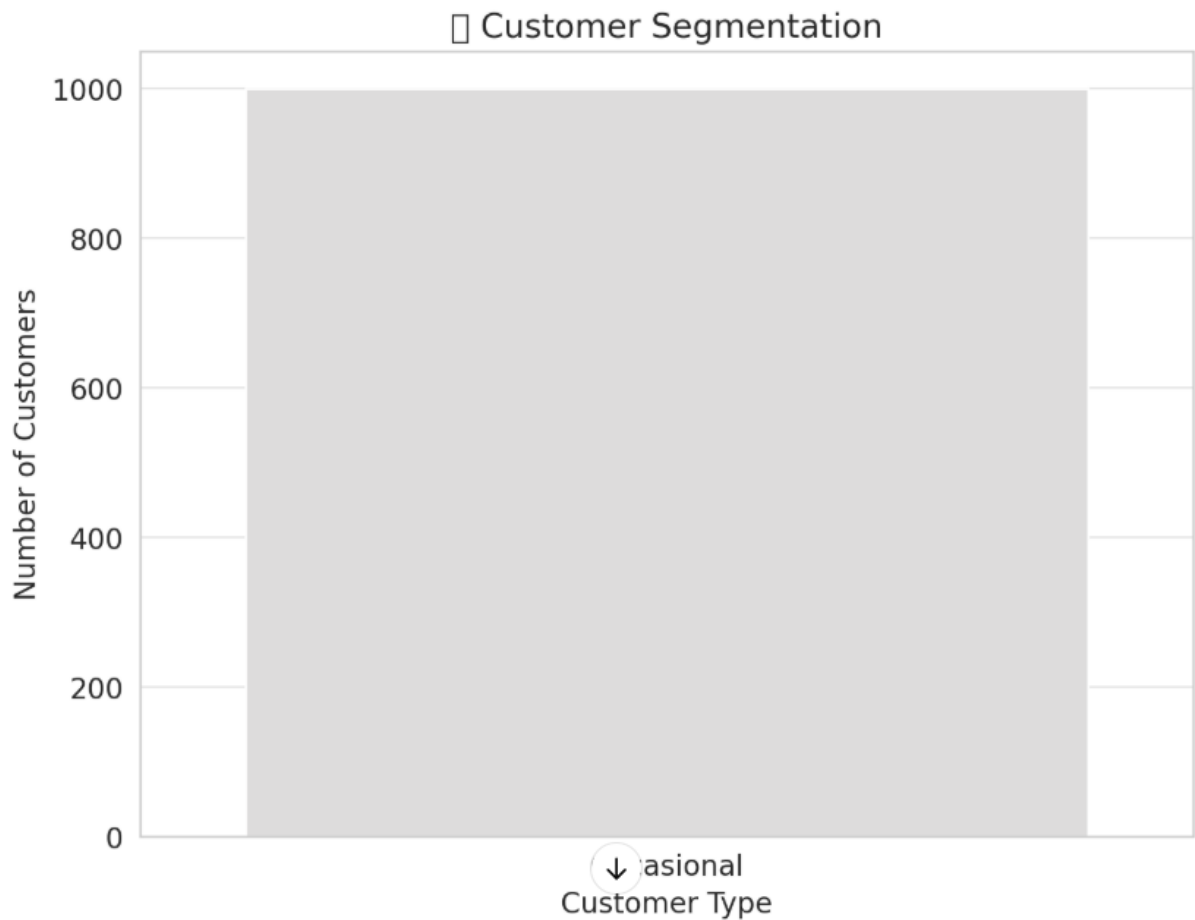
Marketing Focus: Promote high-revenue services more aggressively.

Customer Retention: Collect feedback to improve offerings.

***Insights Reports by Power BI visualizations using line & bar charts:**

Customer Segmentation Insight:

- Majority of customers are **Occasional** (single or few-time bookers).
- There are no significant **Frequent** customers (those with 5+ bookings).
- Business can focus on **loyalty programs** to retain more customers.

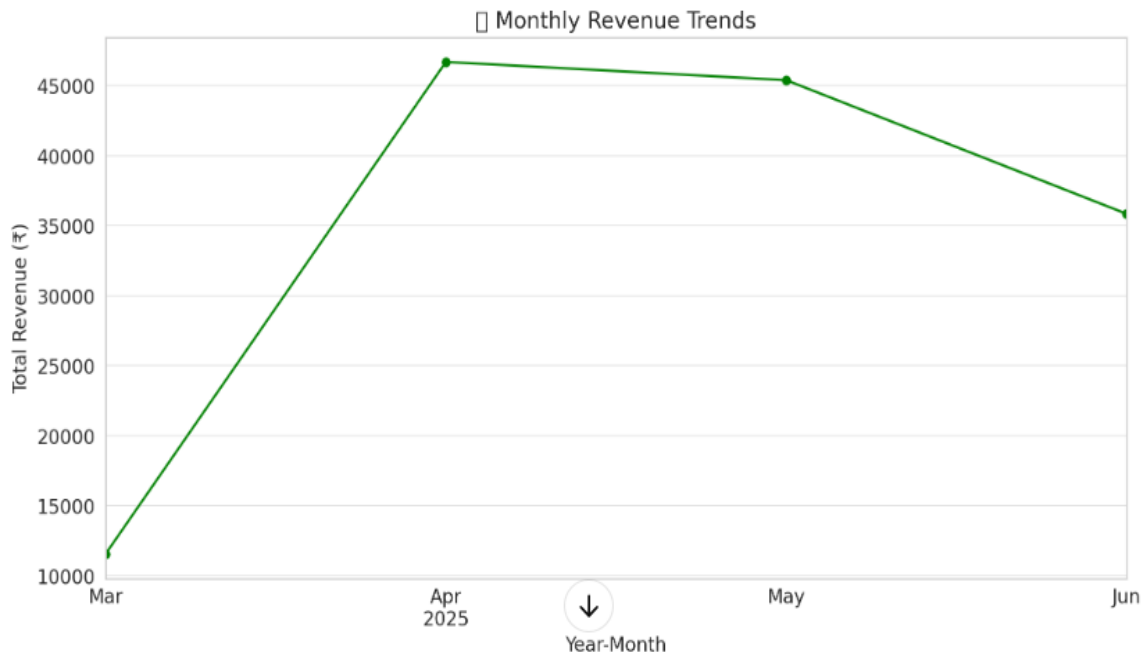


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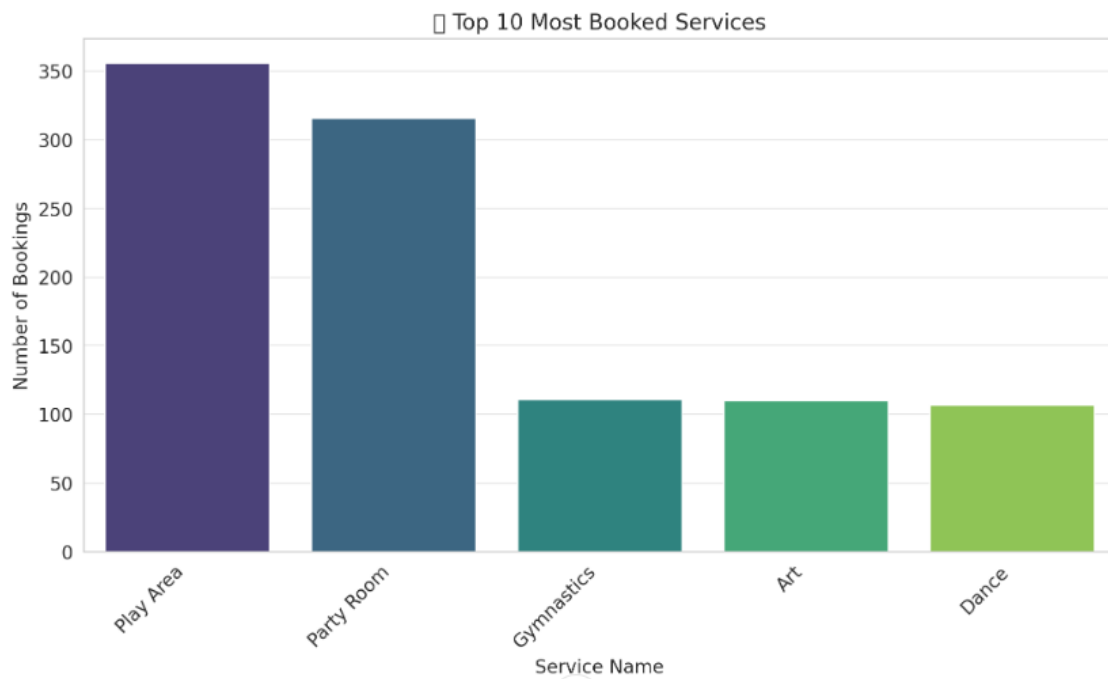
***Revenue Insights:**

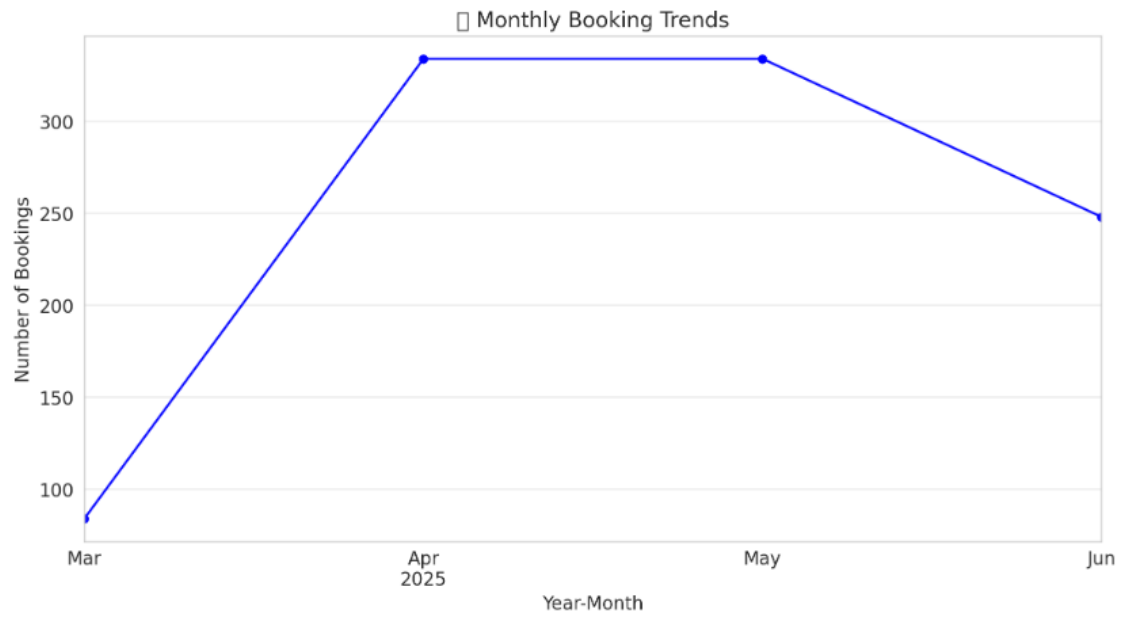
- Revenue fluctuates, aligning with **seasonal demand** seen in bookings.
- Identifying high-revenue months can help with **marketing and promotions**.



Top 5 Most Booked Services:

1. **Play Area** (356 bookings, ₹50,148.64 revenue)
2. **Party Room** (316 bookings, ₹43,762.19 revenue)
3. **Gymnastics, Art, Dance** also contribute significantly.





Monthly Booking Trends Insight:

- The number of bookings fluctuates, indicating possible seasonal demand.
- Further analysis can help identify peak months for revenue generation.